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PROFESSIONAL SUMMARY

Software engineer with 4+ years building production systems end to end. On the backend I work with **Python**, **FastAPI**, and **Node.js** across microservices, REST APIs, and **PostgreSQL**. In AI I build multi-agent systems with **LangGraph** and **LangChain**, **RAG** pipelines on **Pinecone**, and **MCP** servers, backed by LLM evals and cost and latency telemetry. On the frontend I ship **React** and Next.js experiences including human-in-the-loop dashboards. I work directly with stakeholders from problem definition through deployment on **Azure** and **AWS**.

TECHNICAL SKILLS

Programming Languages: Python, JavaScript, TypeScript, Java, SQL, C++, HTML5, CSS3, Shell Script.

Backend: Node.js, Express.js, Flask, FastAPI, Pydantic, Spring Boot, Microservices, REST APIs, API Design, gRPC, System Design, Event-Driven Architecture, OAuth2.0/JWT, Unit Testing (PyTest, Jest), Hibernate, JPA.

Frontend: ReactJS, Vue.js, AngularJS, HTML5, CSS3, Bootstrap, Next.js.

AI & LLM Systems: LangGraph, LangChain, LangSmith, OpenAI, Claude API, RAG Pipelines, Multi-Agent Orchestration, MCP (Model Context Protocol), A2A, Prompt Engineering, Vector Databases (Pinecone, PGVector, Weaviate), Structured Outputs, Model Evaluation & Evals Framework.

Databases: PostgreSQL, MySQL, MongoDB, Redis, DynamoDB, Vector Databases (Pinecone, Weaviate), NoSQL.

Cloud & Infrastructure: AWS (EC2, S3, Lambda, SQS, SNS, CloudWatch, DynamoDB), Azure (Azure OpenAI, Azure AI Foundry, Azure Document Intelligence, Azure AI Search, AKS, Azure Monitor), Docker, Kubernetes, Jenkins, GitHub Actions, CI/CD, Git, Agile.

Monitoring & Observability: LangSmith, Prometheus, Grafana, Datadog, CloudWatch, Distributed Tracing.

EXPERIENCE

AI Software Engineer | Geico | San Jose, CA

Jul 2024 - Present

- Own a production system end to end, with **Python/FastAPI** backend services, **React** dashboards, and a **LangGraph** multi-agent core on **Azure OpenAI** processing 1K+ customer service emails daily at 92% accuracy.
- Built the backend services in **Python** and **FastAPI**, with **REST APIs** orchestrating intent classification, retrieval, and response generation across specialized agents, with A2A-style message passing and state handoffs for reliable agent composition across microservices in distributed systems.
- Designed **PostgreSQL** schemas and stored procedures for conversation state management under high concurrency.
- Built a production **RAG** pipeline with **LangChain**, **LangGraph**, and **Pinecone**, grounding customer email responses in enterprise knowledge bases and cutting average response time from 4 hours to 12 minutes.
- Built **MCP** servers exposing enterprise knowledge bases and internal APIs as context-aware tools, so agents can retrieve customer data, query campaign history, and execute operations programmatically.
- Shipped a comprehensive LLM evaluation framework with **LangSmith**, running automated regression evals over a 200+ case golden dataset and catching 15+ production edge cases before customer impact.
- Made deliberate model and cost decisions, evaluating GPT-4o against GPT-4o mini and routing classification to the smaller model for latency and generation to the larger for quality, batching low-volume workloads, and tracking per-run token spend.
- Deployed an **Azure AI** document pipeline with Azure AI Foundry, Document Intelligence, and Azure OpenAI, running classify-extract-remediate workflows with confidence-based human review routing and per-run cost and latency telemetry.
- Partnered with customer-service stakeholders to build the **React** and Material-UI human-in-the-loop dashboards for agent performance monitoring, low-confidence review queues, and real-time approval workflows, cutting triage turnaround time by 40%.
- Run production with end-to-end observability across **LangSmith**, **Grafana**, and Azure Monitor for pipeline visibility, debugging, and incident response.
- Ship through CI/CD with **GitHub Actions** and Azure DevOps, with automated testing, containerized builds, and production deploys at 85%+ test coverage with PyTest and Jest.
- Work AI-native, using Claude Code and Cursor to cut boilerplate by 40% while keeping code quality high.

Software Analyst | Deloitte | Hyderabad, India

Aug 2021 - Aug 2022

- Architected scalable backend infrastructure for enterprise clients supporting 50+ microservices handling millions of daily

transactions, designing **Node.js** orchestration layer with request batching, retry logic, and circuit breakers for resilience under high load.

- Collaborated on Deloitte's SaaS AI Framework, leveraging **Datadog** to monitor 50+ services, evaluate AI ethics, protect customer privacy, and reduce AI bias by 30%, while communicating insights to client stakeholders and cross-functional teams.
- Shipped a **Node.js** API gateway and **Python FastAPI** services with OAuth2.0/JWT authentication, rate limiting, and structured logging for secure communication across distributed services, reducing API latency by 25%.
- Developed **Vue.js** and **ReactJS** real-time monitoring dashboards with Redux state management for service health visualization and production troubleshooting, reducing incident response time by 30-35%.
- Wrote the Swagger/OpenAPI 3.0 contracts for scalable **REST APIs** that 50+ microservices built against, enabling contract-driven development across distributed teams.
- Optimized database performance across **PostgreSQL**, **MongoDB**, and **Redis**, implementing caching strategies, query optimization, and schema design, reducing database load by 30% and improving throughput by 40%.
- Analyzed client Model Risk Management projects in healthcare, evaluating model performance, mitigating risks, and upgrading algorithm efficiency by 25%.

Software Engineer | Virtutech | Hyderabad, India

May 2020 - Jul 2021

- Built the core backend services in **Node.js** and **Python** (FastAPI): customer-facing RESTful APIs and third-party integrations for real-time data exchange, improving data retrieval efficiency by 50%.
- Automated CI/CD with AWS CodePipeline, CodeBuild, and Docker, cutting deployment time by 40% and sustaining 99.9%.
- Shipped reusable UI components and cut frontend render times with lazy loading, code splitting, and memoization, reducing page load times by 30%.
- Hardened authentication with JWT and OAuth 2.0 across login and session flows, tightening application security without breaking the user experience.
- Built asynchronous event-driven workflows using message queues and background job processing, enabling reliable handling of high-volume transactions and notifications.
- Optimized SQL and NoSQL database queries, indexing strategies, and caching mechanisms, significantly improving API response times and scalability under peak traffic.
- Containerized applications with Docker and deployed services on AWS infrastructure, ensuring consistency across development, staging, and production environments.

PROJECTS

Multi-Agent Research System: AI-Powered Deep Research Platform

github.com/pssgowtham/multi-agent-research

- Built a multi-agent research platform using **LangGraph**, orchestrating 5 specialized agents with a conditional feedback loop for iterative refinement. Integrated semantic search and **Pinecone** vector memory with similarity-based caching.
- Implemented a **FastAPI** backend with SSE streaming for real-time agent progress, multi-layered safety guards (**LlamaGuard** prompt injection detection, PII filtering). Monitored with **LangSmith** for end-to-end LLM observability.

CodeReviewer AI: Autonomous LLM-Powered Pull Request Review Agent

github.com/pssgowtham/codereviewer

- Built an autonomous code review agent with tool-calling and **MCP** integration, analyzing 500+ PRs/month and delivering context-aware feedback on security, performance, and maintainability, reducing review time by 60%.
- Implemented Tree-sitter AST parsing with RAG (**Pinecone**) for repo-aware insights and deployed via **AWS Lambda** with **DynamoDB** and GitHub webhooks, achieving 88% precision on issue detection.

EDUCATION

Master of Science in Software Engineering | San Jose State University

Aug 2022 - May 2024